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You can use bigger strides in a convolutional layer instead of a max-pooling layer for better accuracy.

This statement is not entirely true. While using larger strides in a convolutional layer can help reduce the spatial resolution of the output feature map, which can lead to a similar effect as max-pooling, it does not necessarily lead to better accuracy.  

Max-pooling layers have been widely used in convolutional neural networks (CNNs) to reduce the spatial resolution of the feature maps, which can help to improve the computational efficiency of the network and reduce overfitting. However, max-pooling also discards some information that may be important for accurate classification.

Using larger strides in a convolutional layer, on the other hand, can also reduce the spatial resolution of the feature maps and allow the network to process more efficiently, but it can also result in loss of information. The choice between using larger strides in convolutional layers versus using max-pooling layers depends on the specific task and architecture of the network.

In some cases, using larger strides in convolutional layers may lead to better accuracy, especially when combined with other techniques such as dilated convolutions or skip connections. However, in other cases, max-pooling may still be a better choice. It is important to experiment with different architectures and hyperparameters to find the best configuration for a given task.



How many conv layers with 3x3 filters would you need to have the same receptive field as a conv layer with 11x11 filters? Assume stride=1 and no padding.

If we want to achieve the same receptive field as a convolutional layer with 11x11 filters using only 3x3 filters, we can calculate the number of layers required by the following formula:  

$$n = (d-1)/2$$

where n is the number of layers needed, and d is the size of the receptive field of the target layer.

For a convolutional layer with 11x11 filters, the receptive field size is 11x11. Therefore, we can calculate the number of layers required as follows:

$$n = (11-1)/2 = 5$$

So, we would need 5 convolutional layers with 3x3 filters to achieve the same receptive field as a convolutional layer with 11x11 filters, assuming stride=1 and no padding. Each 3x3 convolutional layer would have a receptive field of 3x3, and stacking 5 of them in sequence would result in a receptive field of 11x11, which is equivalent to the target layer.

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